IN THE HIGH COURT OF JUSTICE

BUSINESS AND PROPERTY COURTS — CIRCUIT COMMERCIAL COURT

Claim No. BL-2025-000418

PENNINGTON INDUSTRIES LIMITED v TREGARTH BEARINGS LIMITED

METALLURGICAL FAILURE-ANALYSIS REPORT

Prepared by Dr Adrian Renwick CEng FIMMM — Argent Metallurgical Services

Report dated 16 December 2024

1. Instructions and scope

1. I was instructed by Whitfield Carter LLP, solicitors for Pennington Industries Limited, to investigate the cause of premature failures of TB-7208-HP angular-contact spindle bearings supplied by Tregarth Bearings Limited and incorporated into PX-180 and PX-220 centrifugal pumps. I examined a sample of 24 failed bearings drawn from the five delivered batches.

2. Summary of conclusions

2. The sampled bearings failed by sub-surface initiated rolling-contact fatigue (spalling) of the inner-ring raceway. The raceway surface hardness of the sampled bearings ranged from 52 to 55 HRC, materially below the specified minimum of 60 HRC. The hardness deficiency is consistent with an incorrect or incomplete through-hardening heat-treatment of the 100Cr6 bearing steel. The reduced hardness substantially lowered the fatigue resistance and rating life of the bearings, which failed at a mean of approximately 4,100 operating hours against a specified L10 life of 30,000 hours.

3. Methodology

3. Each bearing was visually examined, photographed and sectioned. Raceway hardness was measured by the Rockwell C method (HRC) at five locations per ring in accordance with BS EN ISO 6508-1, and corroborated by micro-hardness traverse (HV1). Microstructure was examined by optical and scanning electron microscopy on etched sections, and fracture surfaces were examined for fatigue features.

4. Findings

4. Hardness. The mean raceway hardness of the 24 sampled inner rings was 53.4 HRC (range 52–55 HRC). No sampled ring met the 60 HRC requirement. The micro-hardness traverse showed a uniform through-section hardness with no evidence of the higher near-surface hardness that correct hardening and tempering would produce.

5. Microstructure. The microstructure comprised tempered martensite with retained, partially spheroidised carbides, consistent with under-austenitisation or an over-long temper, either of which would depress hardness. There was no evidence of grinding burn or decarburisation attributable to subsequent processing by Pennington.

6. Fracture features. The raceway spalls exhibited classic sub-surface fatigue origins below the Hertzian contact zone, with no evidence of surface-origin distress such as denting, debris bruising or smearing. This pattern is characteristic of material/hardness inadequacy rather than misapplication.

5. Consideration of alternative causes

7. I considered whether axial overload, excessive operating temperature or inadequate lubrication could account for the failures. Each of those mechanisms produces characteristic surface-origin distress (e.g. skidding/smearing, lubricant degradation staining, or surface-origin spalling), which I did not observe. The consistent sub-surface fatigue pattern across bearings from different batches and pumps points to an

inherent material/hardness deficiency, and I am able to exclude the alternative causes as the primary mechanism.

6. Effect on rating life

8. Bearing fatigue life is strongly dependent on raceway hardness. A reduction from 60 HRC to ~53 HRC is broadly consistent with a fall in fatigue life to a small fraction of the rated value, which accords with the observed mean life of ~4,100 hours against the specified 30,000-hour L10.

7. Declaration

9. I understand that my overriding duty is to the Court and I have complied with that duty. I confirm that this report contains all matters within my knowledge or expertise relevant to the issues on which I have expressed an opinion, and that I have drawn attention to any matter which would affect the validity of that opinion. I believe that the facts I have stated in this report are true and that the opinions I have expressed are correct.

Dr Adrian Renwick CEng FIMMM
Argent Metallurgical Services — 16 December 2024